Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
7	*(a)*	(i)	Collisions / deceleration at metal target	1			1		
		(ii)	$\frac{100 \times 10^{-3}}{1.6 \times 10^{-19}}$ (1) 6.25×10^{17} with no units (1)		2		2	2	
		(iii)	Use of $v^2 = u^2 + 2ax$ and $u = 0$ (1) $\frac{(1.5 \times 10^8)^2}{2 \times 0.03}$ (1) 3.75×10^{17} [m s^{-2}] (1)		3		3	3	
		(iv)	Greater current (gives more intensity) / more X-rays / photons Ignore reference to power and distance (1) Greater pd (gives larger photon energies) (1)	1	1		2		
	(b)		Disadvantage high exposure to radiation / expensive / longer <u>exposure time</u> (1) Advantage 3D images / show range of tissue types (1) **don't accept** more detailed image	2			2		
	(c)		1.76×10^6 (1) 1.20×10^6 (1) ignore s.f. and units yes there will be a large reflection / will reflect / 0.036 or 3.6% (1) Acoustic impedance (significantly) different (1)			4	4	2	
	(d)	(i)	Cause hydrogen nuclei / protons (1) To flip / precess / change orientation / parallel to antiparallel (1) **don't accept** nuclei moving up energy levels	2			2		
		(ii)	$\frac{64}{x} = \frac{1.5}{1.2}$ (1) x = 51.2 MHz **with units** (1)		2		2	2	
	(e)	(i)	$55 \times 10^{-3} \times 20$ = 1.1 Sv answer **with units**		1		1	1	
		(ii)	Less than as (a lot) less ionising / interaction with matter /less biological effectiveness – **don't accept** beta is smaller or mass is less			1	1		
			Question 7 total	**6**	**9**	**5**	**20**	**10**	**0**